SUPPLY CHAIN INVENTORY ANALYSIS

Data Cleaning & Pivot Table Documentation

1. Dataset Overview

The raw dataset contains 1,000 inventory records with the following columns: Product ID, Product Name, Category, Warehouse, Location, Quantity, Price, Supplier, Status, and Last Restocked. Before analysis, the dataset was cleaned to correct formatting inconsistencies and fill in missing values, and several calculated columns were added to support the pivot table analysis.

2. Data Cleaning Steps

2.1 Product Name & Category (Formatting)

Both columns contained inconsistent spacing and inconsistent capitalization (e.g." gadget y ", "FURNITURE"). These were standardized using TRIM to remove extra spaces and PROPER to apply consistent capitalization.

```
=PROPER(TRIM(B2))
```

2.2 Quantity (Text Values)

The Quantity column contained a text entry ("two hundred") mixed in with numeric values. This was converted to its numeric equivalent so the column could be used in calculations.

```
=IF(D2="two hundred", 200, D2)
```

2.3 Last Restocked (Inconsistent Date Formats)

This column mixed real Excel date values with text dates (e.g. "20/12/2022"), which caused errors in date-based calculations. A formula was used to check whether the cell already held a real date, and if not, to rebuild the date manually from its text position (day / month / year) rather than relying on DATEVALUE, which produced #VALUE! errors due to regional date settings.

```
=IF(ISNUMBER(J2), J2, DATE(VALUE(RIGHT(J2,4)), VALUE(MID(J2,4,2)),
VALUE(LEFT(J2,2))))
```

2.4 Missing Price and Quantity Values

207 cells in Price and 158 cells in Quantity were blank. These were filled by randomly sampling from the actual values already present for that Category, rather than using a single repeated value, so the filled-in data reflects the natural variation in the rest of the dataset.

2.5 Missing Status Values

Some rows had a blank Status field, which caused a "(blank)" category to appear in pivot tables. These blanks were identified and resolved so that every row has a valid stock status (In Stock, Low Stock, or Out of Stock).

3. Additional Calculated Columns

3.1 Inventory Value

Added to quantify how much stock value each product represents. Calculated as Quantity multiplied by Price.

```
=Quantity * Price
```

3.2 Days Since Restock

Added to measure how long it has been since each product was last restocked. TODAY() is used instead of a fixed date so the column recalculates automatically every time the file is opened, without needing manual updates.

```
=TODAY() - Last Restocked
```

Note: this column must be formatted as Number (not Date) otherwise Excel may display the result as a date (e.g. 5/6/1903) instead of a day count.

4. Pivot Tables Created

Five pivot tables were built from the cleaned dataset (including the calculated columns above), each answering one of the core business questions.

4.1 Inventory Value by Category / Warehouse

Field	Placement
Category (or Warehouse)	Rows
Inventory Value	Values — Sum

4.2 Stock Status by Location

Field	Placement
Location	Rows
Status	Columns
Product ID	Values — Count

4.3 Out-of-Stock Rate by Supplier

Field	Placement
Supplier	Rows
Status	Columns

Field	Placement
Product ID	Values: Count, shown as % of Row Total

Setting the Value Field to “Show Values As → % of Row Total” converts the raw counts into each supplier's out-of-stock rate automatically.

4.4 Days Since Last Restock by Product

Field	Placement
Product Name	Rows
Days Since Restock	Values: Average

4.5 Products by Stock Status and Location (filterable)

Field	Placement
Product Name	Rows
Status	Columns
Location	Filter
Product ID	Values: Count

This pivot table identifies which specific products are Low Stock or Out of Stock, and allows filtering down to a single aisle at a time using the Location filter.

5. Slicers

Two slicers were added to allow interactive filtering across the dashboard's pivot tables and charts:

- Warehouse: filters the dashboard down to a single warehouse.
- Status: isolates Out of Stock or Low Stock items across all connected visuals.

Slicers were connected to multiple pivot tables via Report Connections, so selecting a value updates every linked table and chart at once.

6. Insights

6.1 Inventory Value by Category / Warehouse

Capital is spread almost evenly across categories, but Clothing and Electronics hold noticeably more inventory value than Furniture and Toys roughly $80,000–$120,000 more each suggesting these two categories are being prioritized for inventory investment.

Warehouse 1 carries more inventory value than Warehouse 2 and Warehouse 3 individually (about $100K–110K more than each). This concentrates risk in one location: a disruption at Warehouse 1 would affect a larger share of company inventory at once. Warehouse 2 and Warehouse 3 are almost identical, showing balanced distribution between those two.

6.2 Stock Status by Location

Shortages are not spread evenly across the warehouse, they are concentrated in two specific products. Widget B is short (Out of Stock exceeding In Stock) in 4 of 5 aisles (Aisle 1, 2, 3, and 5), most severely in Aisle 5. Gadget X is short in 2 of 5 aisles (Aisle 3 and Aisle 5), most severely in Aisle 3. All other products (Gadget Y, Gadget Z, Widget A, Widget C) remain healthy across every aisle. Because the pattern repeats across different aisles, the issue is most likely tied to how these two products are supplied or restocked, rather than to any single location.

6.3 Out-of-Stock Rate by Supplier

All four suppliers perform fairly similarly, but Supplier D is the weakest, with the highest out-of-stock rate at 36.02%. Supplier B is the most reliable, with the highest in-stock rate at 35.42% and one of the lowest out-of-stock rates. Supplier C carries the largest share of items sitting in “low stock” (36.21%), meaning many of its products are close to running out even though few are fully out of stock yet.

6.4 Days Since Last Restock

Every product has gone over 1,280 days (more than 3 years) since its last restock. This is a company-wide restocking gap rather than a single product's issue. Widget C has gone the longest without a restock (1,296.50 days average) and is the most overdue, though Gadget X, at 1,283.63 days, is only marginally better.

7. Recommendations

7.1 Inventory Value by Category / Warehouse

- Confirm that inventory spending on Clothing and Electronics matches actual demand for those categories.
- Review the risk of Warehouse 1 holding more inventory value than the other two warehouses combined, and consider rebalancing stock across locations.

7.2 Stock Status by Location

- Prioritize restocking Widget B and Gadget X, starting with Aisle 5 (Widget B) and Aisle 3 (Gadget X), where shortages are most severe.
- Investigate the supply or restocking process for these two specific products, since the shortage pattern repeats across multiple aisles.

7.3 Out-of-Stock Rate by Supplier

- Review Supplier D's reliability and consider sourcing a backup supplier.
- Use Supplier B's performance as a benchmark when evaluating the other three suppliers.
- Monitor Supplier C's high low-stock rate before it turns into an out-of-stock problem.

7.4 Days Since Last Restock

- Restock Widget C first, as it is the most overdue product.
- Set a maximum allowed restock interval company-wide to prevent products from going years without restocking.
- Build a tracking or alert system to flag overdue items automatically going forward.